Md. Abdullah Al-Fahim

Civil Engineering Researcher | Computational Geomechanics

fahim.civilworks@gmail.com | Dhaka, Bangladesh

linkedin.com/in/fahim-abdullah-f132 | fahim9646.github.io | ORCID: 0009-0003-2483-4536 | Google Scholar

ACADEMIC CREDENTIAL

Rajshahi University of Engineering & Technology (RUET) **2019 – 2024**

Bachelor of Science in Civil Engineering *Rajshahi, Bangladesh*

- **CGPA:** 3.06 / 4.00

STANDARDIZED TEST SCORE

IELTS Academic **Oct. 2026**

Overall Band Score: ***8.0*** *CEFR Level C1*

- **Listening:** 8.5 | **Reading:** 9.0 | **Writing:** 7.5 | **Speaking:** 7.0

RESEARCH EXPERIENCE

Undergraduate Thesis **2023 – 2024**

Dept. of Civil Engineering, RUET *Supervisor: Prof. Dr.Md. Mahmud Sazzad*

"Influence of Inherent Anisotropy of Soil on the Behavior of Sand-Concrete Interface"

- Built a custom 2D **GPU-accelerated DEM engine** in **Python**/**PyTorch** to simulate direct shear of sand and sand–concrete interfaces.
- Calibrated against **48 direct shear tests** (ASTM D3080): two sands, four bedding angles, three normal stresses.
- Used a **calibrate-once protocol**: parameters tuned at one bedding angle, then frozen, so all other angles are blind predictions.
- Analysed how bedding-plane anisotropy transfers from the sand–sand to the sand–concrete interface using stress–strain response and fabric tensors.

Barriers to Green Buildings in Bangladesh **2024**

Survey and Analytic Hierarchy Process *RUET*

- Survey and Analytic Hierarchy Process analysis of the hindrances to green building construction (n = 87 responses; consistency ratio 0.011); **government involvement ranked first (25.0%)**. First-author conference paper (ICCESD 2024).

Finite Element Analysis of a Push-Out Test **Ongoing**

Numerical modelling (FEM) *Manuscript in preparation*

- **Scope: numerical modelling** of the push-out test using the finite element method.

Critical Strategies for Construction Waste Management **Ongoing**

Circular economy, practitioner survey *Manuscript in preparation*

- Second-author manuscript on critical construction waste management strategies from a circular economy perspective, based on a practitioner survey in Bangladesh; responsible for the statistical analysis of the survey data.

PUBLICATIONS

Journal Articles

- **Al-Fahim, M.A.**, Sazzad, M.M., Shafi, M.S.M. *Transmission and Attenuation of Bedding-Plane Anisotropy from Sand to Sand-Concrete Interfaces: A Discrete Element Study.* **Under Review.**

Conference Papers

- **Al-Fahim, M.A.**, Shahriar, S. *Hindrances in Constructing Green Buildings in Bangladesh: A Comprehensive Study.* 7th International Conference on Civil Engineering for Sustainable Development (**ICCESD 2024**), KUET, Khulna, Bangladesh.

TECHNICAL SKILLS

Computational: Python (PyTorch, custom DEM/FEM code), R, ABAQUS

Laboratory: direct shear testing (ASTM D3080), particle-size analysis, concrete mix design (ACI 211), relative density testing, aggregate property testing (ACV, LA abrasion, Fineness Modulus, Specific Gravity)

Methods: calibration and validation of numerical models, survey design, Analytic Hierarchy Process

Design & Productivity: AutoCAD (2D), SketchUp, Lumion; MS Word, PowerPoint, Excel

PROFESSIONAL EXPERIENCE

RANCON Engineering & Consultants Ltd. **Nov. 2025 – Present**

Executive (QC-PMO) *Tejgaon, Dhaka*

- Inspect structural, architectural and finishing works and incoming materials against approved drawings, specifications, BOQ and BNBC; resolve non-conformities with contractors and keep QA/QC records (checklists, NCRs, corrective actions).
- Independently developed a live internal **portfolio-tracking dashboard** (Python/Flask) integrating project schedules with progress, dependency and financial reporting across concurrent developments.
- Developed and piloted **RANCON QMS**, a web-based digital quality management system.

Mir Cement Ltd. **Jan. 2025 – Nov. 2025**

Engineer (Civil) *Rupganj, Narayanganj*

- Developed and monitored project schedules and prepared cost estimates, budgets and resource plans.
- Designed and optimised concrete mixes per ACI 211.R and verified raw materials and finished concrete against quality standards.

PRESENTATIONS & ACTIVITIES

- Poster Presentation – Retrofitting Old Toilets for Sustainable Bangladesh (**WaterAid Toilet Conference, 2025**)
- Team Event – Game of Trusses (**Ceenovation, IUT, 2024**)
- Poster Presentation – Sustainable Microgrids: Enhancing Campus Energy Resilience (**Technofair, RUET, 2023**)

EXTRA-CURRICULAR ACTIVITIES

- **Achievements:** Merit-based scholarships from Sylhet Board
  - Secondary School Certificate: Merit-Based Scholarship
  - Higher Secondary School Certificate: Merit-Based Scholarship
- **Sports:** Champion in football and basketball; captained the house volleyball team (college)
- **Leadership:** Assistant House Prefect (college)
- **Stage competitions:** Participated in inter-house extempore speech and spelling bee competitions